GAME DESIGN DOCUMENT · PRODUCT & PRODUCTION BIBLE

# MNKISWAN

## COZY WORLD

A tiny living world, built around two best friends.

FULL GDD · VERSION 1.0 · 22 OCTOBER 2026



PC / STEAM · THIRD-PERSON 3D · STYLIZED MINIATURE WORLD · GODOT 4.7.2

# Document Control

This is the source-of-truth design and production document for **MNKISWAN: Cozy World**. It is written for game design, programming, character and environment art, animation, audio, QA, marketing, and future collaborators. It distinguishes the existing proof-of-concept from intended production requirements. Numbers in the economy, budget, schedule, and price sections are hypotheses to test, not guarantees.

| Field | Decision |
|---|---|
| Document version | 1.0 — foundational full GDD |
| Prepared | 22 October 2026 |
| Product | MNKISWAN: Cozy World |
| Genre | Third-person cozy life simulation, exploration, light farming and shop management |
| First platform | Windows PC; Steam distribution intended |
| Engine target | Godot 4.7.2 stable [1] |
| Visual style | Stylized 3D miniature diorama, soft forms, tactile materials and expressive characters |
| Player mode | Single-player MVP; no online dependency |
| Core protagonists | Mnki and Swan; all core story and systems are built around them |
| Current status | Early systems prototype using primitive meshes; final character assets, polished animations, robust persistence and production polish remain to be built |

**Scope labels. MVP** means required for the first coherent commercial release or its vertical slice. **Stretch** means implement only after MVP stability. **Post-launch** means not promised for the first release. **Open decision** means the proposed default should be validated before production lock.

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# 01. Executive Summary and Product Definition

## High concept

**MNKISWAN: Cozy World** is a third-person 3D single-player cozy life simulation in a miniature woodland settlement called Tree Village. The player explores as Mnki while Swan travels as an active companion. Together they gather fruit and berries, grow crops, fish, care for domestic animals, help villagers, decorate their shared two-storey treehouse, and develop a small fruit stall into a valued part of the local economy.

The game should feel like inhabiting a lovingly built diorama. The player does not conquer a world; they become part of it. Every short session offers a clear, low-pressure action, and meaningful effort results in a visible change: a fuller stall, a repaired path, a healthier garden, a new route, a keepsake in the treehouse or a villager who now feels like a friend.

> **North-star promise:** Build a little world worth returning to — together.

## One-sentence pitch

Explore a hand-crafted miniature Tree Village with Mnki and Swan, grow and gather beautiful things, run a small fruit stall, care for animals, and help neighbors make their home more welcoming.

## Product pillars at a glance

1. **Character-led:** Mnki and Swan are the emotional center, not decorative mascots pasted onto a generic simulator.
2. **Cozy agency:** choose the activity and pace; no hunger bar, punishment timer or permanent crop failure.
3. **Visible consequence:** the environment, home, stall, NPC routines and dialogue react to player progress.
4. **Tactile miniature world:** rounded shapes, crafted scale cues, landmarks, warm light and calm audio.
5. **A meaningful small economy:** gathering, farming and animal care feed into the fruit stall, orders and upgrades.
6. **Focused production:** make a compact village feel complete before enlarging it.

## The intended first release

| Dimension | Target |
|---|---|
| Platform | Windows PC first; Steam release planned |
| Perspective | Third-person 3D camera behind and slightly above Mnki |
| Visual language | Stylized 3D miniature diorama |
| Play structure | Single-player, exploration hub with short loops and compact adjoining regions |
| Character structure | Mnki controlled by player; Swan follows, reacts and participates in story. Character switching is a stretch decision. |
| Main progression | Village restoration, fruit-stall improvements, home personalization, collection journal and relationship milestones |

| Dimension | Target |
| --- | --- |
| Content target | 13 named locations at compact scale; 6–8 named villagers; 10–15 quests; 4–6 crops; 3 animal species; 3–5 hours of first-release content as an initial target to validate |
| Commercial model | Premium one-time purchase; no paid energy or premium currency |

## What the game is not

It is not a punishing survival simulator, a combat-first RPG, a sprawling procedural sandbox, a multiplayer economy or a large-scale colony management game. Online multiplayer, complex romance simulation, deep crafting, hundreds of NPCs and a mobile launch in parallel are outside the MVP. These features multiply cost without proving that the core loop works.

## Scope principles

- Each named location must have a distinct mood and gameplay purpose, but not each needs a custom art set or separate large level.
- Mnki and Swan’s canonical designs must be approved before custom modeling and animation are scaled up.
- The fruit stall is the signature progression system and deserves more polish than a decorative shop prop.
- All systems that change inventory, coins, quests or world state must be persistent and resistant to duplicate rewards.
- The first goal is a delightful, testable vertical slice, not a large map filled with unfinished interactions.

# 02. Vision, Pillars and Quality Bar

## Pillar A — Cozy agency

Players can spend a session fishing, decorating, helping villagers, collecting fruit, working a crop plot or simply exploring. The world suggests activities without treating them as obligations. Requests do not expire in a way that harms relationships. Missing a day never kills a crop, permanently loses an animal or closes an essential story path.

## Pillar B — Mnki and Swan are the heart

Their signature outfits, silhouette, movement, reactions and supportive exchanges appear in actual play. Swan celebrates a first successful sale, waits politely while Mnki fishes and reacts to a restored park. Mnki investigates discoveries, cares for animals and presents gathered goods. Their personalities should complement one another without making either feel like a passive prop. Dialogue should be warm and can show affection, but the MVP does not require romance meters or a forced romantic storyline.

## Pillar C — The world visibly responds

A harvested bush changes state and later regrows. Crops have visible stages. The shop display fills when goods are sold. Completed community quests repair or decorate a physical space. Animals respond to care. A new home decoration remains exactly where the player placed it after loading a save.

## Pillar D — Miniature-world readability

World spaces are framed like small crafted sets. Large roots, layered canopy forms, curved paths, cozy tree homes and colorful items communicate location identity. The visual richness must never hide the player, the interaction target or navigable routes.

## Pillar E — Economy creates choices

The stall gives activities a purpose, but there is no single optimal grind route. Players choose whether to sell berries, hold fruit for an order, plant a crop, fish, collect flowers or save coins for a home decoration. Every purchase should have a clear benefit.

## Pillar F — Production discipline

Complete one reliable version of every core system before deepening it. One well-tuned fishing loop is better than four unfinished modes; eight memorable villagers are better than fifty empty houses. The project should be modular enough to expand without letting one script own the entire game.

## Quality bar

| Dimension | Acceptance standard |
|---|---|
| Controls | Walking, turning, camera collision and interaction are predictable. No frequent jitter, stuck states or visible camera clipping. |
| Readability | Collectible objects and ready crops are distinguishable from decoration at normal camera distance. Prompts name the action. |
| Feedback | Each pickup, sale, quest update, animal-care action and save gives clear visual and/or audio feedback. |

| Dimension | Acceptance standard |
| --- | --- |
| Persistence | Inventory, coins, quests, crop states, unlocks, shop upgrades and home decor survive save/load. |
| Tone | No jumpscares, shame-based dialogue, punishing survival loops or predatory monetization. |
| Performance | Initial target is 60 FPS at 1080p on a defined mid-range PC, validated through profiling rather than guesswork. |
| Brand fidelity | Mnki and Swan retain canonical clothing, accessories, colors, silhouettes and signature features in every asset. |

# 03. Audience, Positioning and Commercial Intent

## Primary audience

Players who enjoy cozy exploration, light farming, collecting, animal care, decorating, shop progression and character-driven stories. The experience should be understandable to players who are not expert 3D gamers while providing secrets and completion goals for more experienced players.

## Secondary audience

- Fans of Mnki and Swan who want to spend time with the original characters inside an explorable world.
- Players who like collecting fish, fruit, flowers, keepsakes and home decorations.
- Cozy-game creators who can show a charming gameplay moment in a short clip: a chicken response, a fishing catch, a stall upgrade or a restored garden.

## Positioning statement

For players who want a gentle, character-led escape, MNKISWAN: Cozy World is a miniature village-life game where helping neighbors and growing a tiny business steadily transforms a beautiful shared home. Unlike a purely open-ended sandbox, it has a compact narrative arc and visible community restoration. Unlike a management spreadsheet, its economy is experienced through third-person exploration and direct interaction.

## Competitive distinction

Farming alone is not a durable differentiator. The game should be recognizable through four combined traits: (1) the Mnki-and-Swan character identity; (2) a diorama-like miniature world; (3) a tactile fruit-stall business loop; and (4) the visible restoration of a village through small acts of care. Marketing should demonstrate all four with actual gameplay footage.

## Product metrics that matter

| Metric | Question | Test method |
|---|---|---|
| First-session completion | Can a new player understand movement, interaction and the first goal? | Observe five or more players without coaching; note hesitation and errors. |
| Core-loop repetition | Is gather → sell → spend satisfying? | See whether testers repeat the loop voluntarily after tutorial completion. |
| Demo completion | Is the sample clear and appropriately sized? | Measure completion where consented telemetry exists, or use moderated sessions. |
| Wishlist growth | Does the product pitch gain interest? | Review store traffic and wishlist reporting when the Steam page is live. |
| Return intent | Do players know what they want to do next? | Ask them to name their next activity after the demo. |
| Critical defects | Can the player trust the game? | Prioritize save loss, inventory duplication, softlocks and crashes above minor cosmetics. |

# 04. Premise, Lore, Story and Tone

## Player fantasy

“I can wander through a beautiful little world with someone I care about, make useful things with my time, and see the place become kinder because I was here.” The primary emotional rewards are comfort, competence, curiosity, belonging and shared accomplishment.

## Premise — proposed foundation

Mnki and Swan share a two-storey treehouse near the heart of Tree Village. They discover that the village paths, gardens and local trades need renewed care after a long quiet period. The story begins with simple needs: a neighbor needs fruit for a gathering, the pond needs attention, a gardener needs flowers, a bridge needs repair, or the market has empty crates. Each task connects residents and gradually restores the village’s sense of community.

The three guardian statues — **Land, Water and Forest** — introduce a gentle mystery about stewardship. They are not enemies or a combat progression system. Their symbols ask how a community cares for the ground, water and living things. The MVP may reveal one clue per region and resolve the opening arc with a community celebration; a deeper guardian storyline can be saved for later content.

## Themes

- **Friendship as practical care:** listening, helping, sharing work and respecting different preferences.
- **Gentle stewardship:** look after the land, water, plants and animals rather than “conquering” nature.
- **Small work has value:** one garden bed, one repaired path or one fulfilled order can matter to an entire village.
- **Self-expression:** the player chooses a daily routine and makes the shared treehouse feel personal.
- **Curiosity without fear:** secrets reward observation, not punishment.

## Story delivery

Use short conversations, environmental storytelling, item descriptions, journal notes and occasional brief camera-led moments. Main dialogue should be skippable after its first viewing. Important story information must also be discoverable in the journal. Long cutscenes should not interrupt an otherwise calm activity loop.

## Dialogue rules

- Most interaction lines should be one to three sentences, with a longer scene reserved for milestones.
- Give characters distinct speech patterns without caricature or long exposition dumps.
- Prefer sincere humor, specific sensory detail and small emotional reactions.
- Not every villager interaction should start a quest. Some residents simply greet the duo, notice the weather or comment on a local change.
- Ignoring a request does not hurt a relationship. Use “Whenever you have time” instead of guilt-based reminders.
- Mnki and Swan’s bond is expressed through support, shared discovery and optional scenes. The MVP does not require a numeric romance meter or forced romantic framing.

# 05. Canonical Character Bible

## Mnki — canonical identity

**Silhouette:** cute, playful monkey boy with rounded chibi proportions and a clearly readable curled tail. **Head:** tousled dark-brown hair, warm medium-brown skin/fur and large expressive brown eyes. **Signature accessory:** gold curled/swirl headband. **Clothing:** red hoodie, loose black pants, red-and-white sneakers. **Tail:** long brown monkey tail with curled tip, visible from gameplay angles and animated with secondary motion.

### Mnki movement and expression

- **Idle:** gentle weight shifts, blink, a small tail curl and subtle hoodie movement. Idle variations are low-amplitude and not distracting.
- **Walk:** readable foot placement, soft body bounce and tail lag. No sliding feet on flat terrain.
- **Run:** stronger forward lean and more active tail/arm movement, but the game does not become a speed platformer.
- **Jump and land:** compact, joyful movement. Essential routes must not depend on difficult jumps.
- **Gather:** look toward the target, reach, collect, show the item response and return to locomotion.
- **Care action:** bend or kneel at a believable distance from the animal; hands must not clip through the animal body.
- **Shop interaction:** set goods on the counter, arrange a display or inspect a ledger; this is a signature visual moment.
- **Emotions:** happy, curious, thoughtful, proud, surprised and content. Expressions should not become mocking or exaggerated to the point of breaking character.

## Swan — canonical identity

**Silhouette:** swan princess with large white feathered wings and a recognizable white swan hood. **Hair:** long dark-brown hair. **Face:** warm light/medium skin and large expressive brown eyes. **Hood:** white swan hood with orange beak and a small gold crown. **Clothing:** white top/hoodie with a small swan motif, pink pleated skirt, white socks and blue-and-white sneakers. **Wings:** large white feathered wings, correctly attached and visible. Wings must not be omitted in canonical models or key art.

### Swan companion behavior

- Follows at a comfortable trailing or side distance on safe paths. She should not crowd the player or repeatedly enter the camera.
- Waits during fishing, shopping, gathering, house interactions and dialogue. Wait locations must not block doors, targets or narrow bridges.
- Reacts selectively to a first fish, a newly opened path, a village improvement or a completed major quest.
- Suggests activities only occasionally. Reactions have cooldowns and varied lines to avoid chatter spam.
- When navigation fails, she recalculates a safe point, waits or rejoins unobtrusively when out of view. Visible teleporting directly in front of the camera is a defect, not an acceptable normal behavior.
- Reacts to changes in the home and village using short staged comments. Sparse reactions make milestones feel more meaningful.

## Relationship progression

The bond advances through shared milestones and optional story scenes rather than gift grinding. A relationship flag may unlock extra dialogue, a keepsake, a small joint animation or a new home interaction. It must not gate basic farming, selling or exploration. Shared work should feel cooperative: Mnki handles the

physical gathering and exploration loop while Swan encourages, observes, helps organize tasks and participates in scripted moments. Avoid making her a floating hint box; she needs idle activity, reactions and presence.

| Moment | Mnki | Swan | Purpose |
|---|---|---|---|
| First arrival home | Explores the entrance and checks the rooms | Notices a personal detail and points out useful spaces | Establishes the treehouse as a safe hub |
| First sale | Presents gathered goods | Celebrates the earnings and suggests reinvestment | Makes economy feel shared |
| Animal care | Approaches and performs the care action | Demonstrates patience and gives a small reaction | Teaches gentle care |
| Village restoration | Performs hands-on tasks | Encourages neighbors and comments on visible changes | Shows complementary strengths |
| Rare discovery | Investigates the object | Offers a contextual reaction or journal clue | Rewards curiosity |

## Canon integrity checklist

- Mnki retains red hoodie, black pants, red-and-white sneakers, gold curled headband, dark-brown tousled hair and visible brown curled tail.
- Swan retains white swan hood with orange beak and small gold crown, long dark-brown hair, white top, pink pleated skirt, white socks, blue-and-white sneakers and large white wings.
- Optional costumes, if ever introduced, must be clearly optional and preserve recognizable canonical silhouettes.
- Every model, promotional render, portrait, animation test and turnaround must be checked against the canonical reference sheet.

# 06. Supporting Cast and NPC Rules

Keep the named cast small, distinctive and useful. Names below are working labels except the established character Thronn. Rename working roles before final dialogue recording if needed; do not create dozens of NPCs merely to fill a street.

| Character / role | Location | Personality and need | Gameplay function | MVP depth |
|---|---|---|---|---|
| **Thronn — Mnki and Swan's best friend** | Animal farm / Thronn's treehouse | Warm, practical, playful; knows animals and local paths | First animal-care quest, tutorial help and friendship milestone | 2 quest steps, 6–10 short dialogue lines, simple daily schedule |
| Village Elder — working name | Village park / elder's house | Patient and observant; values small acts of care | Main restoration chain; community board or gate access | 3 story beats and a final community event |
| Gardener A — working name | Flower garden | Practical; notices plant health and color balance | Flower collection and seed reward | 2 requests, 4 dialogue states |
| Gardener B — working name | Flower garden | Creative; wants variety and unusual colors | Optional collection and garden decor | 1 collection request |
| Rice farmer — working name | Paddy field | Steady; explains planting and watering | Teaches crop stages and gives first seeds | Tutorial and repeatable order |
| Fisher — working name | Waterfall / ponds | Quiet and observant | Fishing lesson and fish-journal hint | 1 tutorial quest, optional rotating requests |
| Market grocer — working name | Market square | Friendly and business-minded, not greedy | Explains prices, offers and selling | Shop tutorial and upgrade |
| Decor vendor — working name | Market square | Resourceful and detail-focused | Sells a small curated set of home items | One shop UI and limited inventory |
| Colony residents | Colonies 1 and 2 | Distinct homes and local customs | Short errands, produce requests and optional lore | 3–5 reusable resident roles |

## NPC state model

Separate **ScheduleState** (where the NPC intends to be), **ActivityState** (work/idle/sit/walk animation), **DialogueState** (eligible lines and quest flags) and **SafetyState** (idle/recovery if navigation fails). Avoid one giant nested conditional for all NPC logic.

| Time block | Typical behavior | Design reason |
|---|---|---|
| Morning | Leave home, greet nearby residents, start work | Establishes a readable rhythm |
| Midday | Work at market, farm, garden or pond | Makes each role obvious |
| Late afternoon | Move to a secondary point or socialize | Creates variation without heavy simulation |
| Evening | Return home or sit at a social hub | Communicates time passing |
| Navigation failure | Wait, recalculate path, move to a safe nearby point | Prevents an NPC becoming permanently stuck |

# 07. World Layout and All 13 Locations

The thirteen named locations are a world-plan commitment, not thirteen giant levels. Build one compact connected village with short routes and clear sightlines. Reuse terrain, tree kits, props and material palettes while giving each place a unique landmark, gameplay purpose and sound identity.

## Macro layout

The central market and village park are the social/economic hub. The shared treehouse is within a short walk. Nature-gathering regions form a loop around the hub. The waterfall and Sunshine Forest make the exploration route, while the farms, orchard and berry maze feed the shop. The two residential colonies create optional quest density. The entrance statues anchor the theme and navigation.

### Navigation goals

- From the treehouse door, identify the direction of the market in a few seconds.
- The Three Guardian Statues are a recognizable landmark from the main village path.
- A first-time player can reach the fruit stall within roughly one minute of being shown the main path.
- No core activity requires crossing the entire map repeatedly. Use nearby hand-in points, short loops and restored shortcuts.
- Collectibles sit slightly off main routes to reward curiosity without becoming random clutter.
- Every region has a unique silhouette, dominant color cue and signature sound.

## Schematic region plan




This is a conceptual relationship map, not a measured map. Final geometry needs greyboxing, camera tests and navigation baking. Region transitions may be seamless where practical, but a compact transition or scene load is acceptable if it preserves a calm flow.

## 01 — Sunshine Forest: starting clearing

**Mood and landmark:** Fresh morning light, layered green canopies, flower edges and soft dappled shadows. A sunlit opening frames an obvious path toward the waterfall and village.

**Activities:** First movement/camera tutorial, basic interaction and gathering; look for flowers and berries; observe butterflies. The opening area should teach without a long tutorial wall.

**Resources:** Common flowers and berries; optional fallen sticks or seed packets only if a later system uses them.

**Narrative use:** Mnki and Swan arrive together. Swan points out a landmark and lets the player choose whether to walk toward home or explore first.

**Audio:** Birds, leaf movement, breeze and distant water. No loud repeated notification sound.

**Scope:** Required; small, visually confident and performance-friendly. It sets the whole art promise.

## 02 — Waterfall, Bridge and Ponds

**Mood and landmark:** Cool teal water, wet stone, moss, lily pads, sun flecks and a small timber bridge. The waterfall curtain should be visible from the approach. The fishing bank must be broad enough for the camera and companion.

**Activities:** Fishing tutorial, catch collection, pond observation, frog and butterfly sightings. Bridge crossing supports the village route.

**Resources:** Fish; optional reeds or pond flowers if they have a quest or decor use.

**Narrative use:** The fisher teaches the bite cue. A pond-care request can reveal a small Land/Water/Forest clue.

**Audio:** Layered waterfall ambience, small splashes, distant frogs and insects. Reduce ambience under dialogue.

**Scope:** Required. Fishing must have a clear shoreline position and generous feedback.

## 03 — Cherry Berry Maze

**Mood and landmark:** Playful waist-high hedges and rounded berry bushes. Branching paths are short and readable, not a frustrating labyrinth.

**Activities:** Berry gathering, route-finding, small visual secrets and a possible optional timed challenge only after testing. The main route should be visible from nearby landmarks.

**Resources:** Common berries, occasional rare variants and perhaps a hidden home keepsake.

**Narrative use:** A basket request for a market order or village gathering.

**Audio:** Hedge rustle, bird calls and a cheerful collection cue.

**Scope:** Required as a compact gathering loop. Do not generate a procedurally confusing maze.

## 04 — Village Flower Garden

**Mood and landmark:** Colorful flower beds, a small pond, an arched trellis, benches, watering cans and two gardening stations.

**Activities:** Harvest or request flowers, unlock seed packets and learn decorative planting if home gardening is in MVP.

**Resources:** Flower types by color. Seeds should be a reward, not a grind wall.

**Narrative use:** Two gardeners show different values: one grows practical flowers, the other experiments with variety and color.

**Audio:** Bees, watering, birds and calm garden ambience.

**Scope:** Required. Home flower planting can use a simpler system than field crops.

## 05 — Paddy Field and Farm

**Mood and landmark:** Neat plots, water channels, raised bund paths, a small shed and an open view toward the village.

**Activities:** Plant, water, wait by in-game time and harvest four to six crop types eventually. Visual crop states need to communicate progress.

**Resources:** Rice, vegetables and crop bundles. Compost is post-MVP unless it creates a strong reason to play.

**Narrative use:** The farmer grants starter seeds and requests one harvest; a village order teaches crop-to-shop progression.

**Audio:** Water flow, leaves, farm birds and distant work sounds.

**Scope:** Required, but begin with one field section and a small crop roster.

## 06 — Tropical Fruit Forest

**Mood and landmark:** Layered banana leaves, mango canopy, palms/coconuts and warm sun shafts with patches of shade.

**Activities:** Gather ripe fruit, spot fruit by color and position, learn that some sources regrow after a short in-game interval.

**Resources:** Mango, banana and coconut. The first playable slice may begin with mango and berries before adding variants.

**Narrative use:** A fruit basket for a neighbor, new stall inventory and a landmark discovery.

**Audio:** Leaf flutter, fruit pickup and tropical bird calls; avoid a loud generic jungle loop.

**Scope:** Required as the fruit supply route. Tree climbing is not needed.

## 07 — Domestic Animal Farm and Thronn’s Treehouse

**Mood and landmark:** Small barn, fenced paddock, food troughs, straw, animal-safe paths and Thronn’s treehouse. The farm should feel cared for rather than industrial.

**Activities:** Pet, feed and brush animals; collect eggs; optional wool/milk if models, animation and balancing allow.

**Resources:** Eggs for MVP; milk and wool are stretch items. No animal slaughter.

**Narrative use:** Thronn teaches gentle animal care and helps the duo with a small repair or animal-comfort quest. This reinforces their best-friend bond.

**Audio:** Distinct animal calls, straw steps and soft barn ambience.

**Scope:** Required. Cow, sheep and chicken are visible, with two or three interactions each and persistent care state.

## 08 — Entrance and Three Guardian Statues

**Mood and landmark:** Ceremonial path, three distinct stone or wood statues and offerings of plants, stones or water bowls.

**Activities:** Read a short plaque, notice symbols and unlock the main village route.

**Resources:** Not a repetitive harvest area; offerings are quest objects only.

**Narrative use:** Statues embody **Land**, **Water** and **Forest**, anchoring the gentle-stewardship theme.

**Audio:** Wind in an arch, subtle chime on clue discovery.

**Scope:** Required as landmark and story touchpoint; a long guardian quest chain can wait.

## 09 — Village Garden and Park

**Mood and landmark:** Central green, benches, an old tree, child play area, village elder’s home and a circular walking route.

**Activities:** Meet NPCs, receive quests, observe resident routines and see changes after community tasks.

**Resources:** Primarily social hub; a few optional flowers or hidden collectibles.

**Narrative use:** Community board, elder dialogue, milestone gatherings and visible village improvements.

**Audio:** Soft conversational clusters, birds and possibly a fountain. Do not create a noisy crowd wall.

**Scope:** Required. Use a small cast and simple routines.

## 10 — Mnki and Swan’s Two-Storey Treehouse

**Mood and landmark:** A warm shared home with wood grain, woven mats, leaf shadows, handmade trinkets and character keepsakes.

**Ground floor:** Living/sitting area, storage, front entry and display shelf. It should read as a safe shared base.

**Upper floor:** Compact kitchen area and sleeping area. Use stairs or a clear transition; a separated interior is acceptable if it avoids camera problems.

**Activities:** Rest/save, store selected items, display collectibles, place a small set of decorations, review the journal and plan the next day.

**Narrative use:** First arrival, end-of-day reflection, milestones and optional character scenes.

**Audio:** Quiet interior bed, wood steps, subtle kitchen sounds only if cooking enters scope.

**Scope:** Required. Rooms should be compact; do not overbuild the interior before the world loop is proven.

## 11 — Tree People Colony 1

**Mood and landmark:** Intimate residential lane with varied home silhouettes, little gardens, hanging lanterns and distinctive doorways.

**Activities:** Resident quests, visits, hidden garden objects and optional neighborhood requests.

**Narrative use:** One resident requests produce; another needs help finding a keepsake.

**Audio:** Domestic ambience, wind chimes and distant village sounds.

**Scope:** One compact route with two or three named interactions.

## 12 — Tree People Colony 2

**Mood and landmark:** More vertical tree architecture, safe wooden bridges, different roof shapes and hanging plant baskets.

**Activities:** Short exploration loop and requests that combine resources from other regions.

**Narrative use:** A resident reveals a second statue clue or rewards a home item.

**Audio:** Higher canopy wind and birds, distinct from Colony 1 without being louder.

**Scope:** Visually present but compact. The two colonies may share one continuous district in the MVP.

## 13 — Market Square and Mnki & Swan Fruit Stall

**Mood and landmark:** Warm stalls, fruit color blocks, canvas shades, crates, a notice board and open sightlines so characters remain visible.

**Activities:** Sell produce, compare prices, fulfil orders, upgrade the stall, buy seeds and decorations.

**Resources:** Fruits, berries, flowers, fish, eggs and crops only when each category is supported by the current build.

**Narrative use:** Economic tutorial, first stall upgrade and market-day community milestone.

**Audio:** Light market chatter, basket set-downs, coin cue and gentle bustle. Dialogue always stays clear.

**Scope:** Required and high priority. This is the signature system, not a background prop.

# 08. Core Player Experience, Controls and Session Design

## Game mode

Single-player third-person 3D. Mnki is the controlled character and Swan is an active companion. Switching control to Swan is a stretch decision, not a requirement for the first vertical slice; the MVP still gives Swan companion behavior, contextual reactions, dialogue and direct participation in the main story. A shared controller framework can keep future character switching possible without promising it now.

## Default PC controls

| Action | Default binding | Behavior |
|---|---|---|
| Move | WASD / arrow keys | Camera-relative; acceleration and braking are smoothed |
| Look | Mouse | Sensitivity setting; invert-Y option |
| Interact / confirm | E | Contextual action chosen by the interaction scanner |
| Jump / step | Space | Optional traversal and playful movement; not required for essential routes |
| Run | Left Shift | Toggle or hold option; no stamina consumption |
| Inventory | Tab or I | Final default chosen after usability testing; remappable |
| Journal | J | Main objective, side requests, discoveries and item notes |
| Map | M | Region overview, home, market and known quest markers |
| Pause / settings | Esc | Pauses simulation and releases mouse capture |
| Cancel / back | Esc or right mouse | Contextual; destructive actions always require clear confirmation |

## Camera and movement requirements

- Default view sits behind and slightly above Mnki. Test a tunable follow distance around 3.2–5.0 meters depending on the space.
- Camera collision shortens distance smoothly rather than letting walls cover the character. In interiors, adjust pitch or shoulder offset only if tested.
- The character turns toward movement direction with smoothing. Interaction facing is explicit and stable; it must not cause a sudden 180-degree snap.
- Mouse camera input must not fight UI input. When a menu is open, mouse movement belongs to the UI or remains free, not both.
- Target selection, scene transition and Swan recovery must never produce sudden disruptive camera snaps.
- Provide sensitivity, invert-Y, camera shake off/low and reduced camera motion.
- Normalize movement input so diagonal travel is not faster than straight travel.

## Session design

| Session | Likely activity | Desired endpoint |
|---|---|---|
| 5 minutes | Gather a few berries, check the treehouse, greet Swan | A pleasant micro-goal without remembering a long chain |
| 15 minutes | Complete one request, sell goods, buy a decoration | Clear reward and a changed space |
| 30 minutes | Explore another region, fish, care for animals, advance a quest | One new system or place is unlocked |
| 60 minutes | Finish a village milestone, upgrade the stall and rest | Village or relationship makes visible progress |

## Failure and recovery philosophy

There are no game-over screens for ordinary cozy actions. Falling out of bounds returns the player to a recent safe location and keeps all valid inventory. A missed fishing bite costs a few seconds and no item. A blocked companion waits or recovers. A quest can be left and resumed. A full inventory leaves the world item in place and explains why pickup failed. A failed transaction changes neither inventory nor coins.

# 09. Core Gameplay Systems

## 9.1 Third-person exploration

Movement includes acceleration, braking, gravity, grounded movement, slopes, step navigation, collision, gentle jumping and safe fall recovery. Essential objectives must not rely on difficult jumps or precision platforming. The player capsule must fit common doors and paths with comfortable clearance. Keep a last-safe-ground position for recovery. Interaction objects should not require pixel-perfect positioning.

Use separate collision layers or masks for world geometry, player, companion, NPC/animal bodies, interactables, triggers and camera blockers. Decorative foliage should not unexpectedly trap the player. Collision proxies should be simpler than render meshes.

## 9.2 Interaction framework

All plants, characters, stalls, home objects and animals use one shared interaction contract. Suggested methods: `can_interact(actor)`, `get_interaction_prompt(actor)`, `interact(actor)` and an explicit priority value. The controller gathers candidates, removes invalid/disabled/out-of-range targets, ranks the remainder and calls exactly one validated action.

| Rule | Specification |
|---|---|
| Default reach | Start testing around 2.0–2.6 m for hand interactions; fishing and large objects use separately tuned ranges |
| Prompt | Verb + target/outcome: “Gather berries”, “Talk to Swan”, “Sell basket”, “Rest until morning” |
| Target priority | Combine camera alignment, distance, explicit priority and optional line-of-sight; do not choose a nearby target behind a wall automatically |
| Cooldown | Prevent duplicate shop, dialogue, quest or time-advance actions from one key press |
| Validation | Recheck target availability and inventory/state conditions when the input is pressed |
| Feedback | Prompt, animation or mesh-state change, HUD update, and optional sound; never hidden state only |
| Failure response | Explain why the action is unavailable instead of silently failing |

## 9.3 Gathering and regrowth

Gathering nodes have a stable item ID, yield, regrowth rule, availability state, interaction animation and optional respawn time. Use predictable yields in the MVP so the economy is understandable. Rare variants can reward observation and location discovery rather than punitive randomness.

- Berry bushes visibly change to a picked state and regrow after sleeping or a configured number of in-game hours.
- Mango trees use ground-level interaction points. Climbing is not required. A ripe fruit silhouette/color state signals readiness.
- Decorative flowers stay in the world; collectible flowers use a distinct model or clear affordance.
- When inventory is full, do not delete the source object.
- The journal can record first discovery, species and location without requiring 100% completion.

## 9.4 Farming

The farming loop is deliberately small: choose a plot, choose a seed, plant, water, allow in-game time to pass, then harvest. Crops use shared logic with data-driven growth stages. The MVP begins with 4–6 crop types. If a player ignores the game, a crop does not die; growth can pause until the player returns or watering resumes.

| State | Visual representation | Player action | Persistent data |
|---|---|---|---|
| Empty | Prepared soil | Plant selected seed | Plot ID; crop empty |
| Planted | Seed mound and small marker | Water and wait | Crop ID; planted day; stage |
| Growing | 2–3 visibly different stages | Water/inspect | Stage; watered flag/day; elapsed game days |
| Ready | Clear mature fruit/leaf shape | Harvest | Ready state and consumed yield |
| Recovery | Empty or short regrowth state | Replant when ready | Regrowth timer or next eligible day |

Proposed crop set: rice, leafy vegetable, tomato-like vegetable, flower crop and a simple root crop, with one optional sixth crop only if it serves an order or economy need. Each CropDefinition sets seed item ID, harvest item ID, days to maturity, water requirement, yield and visual stage assets. Exact grow durations must be tuned in playtests and demonstrated during a normal demo session.

## 9.5 Fishing

Fishing is a calm timing activity, not a reflex test. The player stands in a validated fishing zone, starts fishing, receives a visual/audio bite cue and confirms within a generous window. MVP uses one two-step loop: wait for bite, respond to cue, play catch animation, add fish to inventory. Multiple fish species can be added by location or time-of-day later; do not build a complex rarity system before the first loop works.

- If the shoreline is crowded, Swan moves to a wait point and does not block the rod/camera.
- Use bobber motion, splash, a sound cue and UI reinforcement. Audio is never the only signal.
- Guarantee an ordinary catch early in the first fishing session after a reasonable number of attempts.
- A missed bite costs seconds only; it never destroys a rare item or coins.
- The journal records fish name, location and a short description.
- Include an easier timing option if the final mechanic requires a narrow reaction window.

## 9.6 Animal care

Animals are friendly neighbors, not industrial production units. Each animal stores last-fed day, last-groomed day and a capped comfort/affection state. The MVP includes a cow, sheep and chicken; petting and feeding are available for all three. Chicken eggs can be supported as the first output. Milk and wool are stretch items if animation and balance are affordable. No animal slaughter or animal death is included.

| Action | Immediate response | Longer-term result | Comfort rule |
|---|---|---|---|
| Pet | Animal turns, blinks or vocalizes | Small affection response or journal note | Repeated input may get a short cooldown but no penalty |
| Feed | Item consumed and eating animation | Fed state for that in-game day | Explain accepted food before consumption |
| Brush/groom | Brief grooming animation | Comfort state or quest progress | Short and skippable after first time |

| Action | Immediate response | Longer-term result | Comfort rule |
|---|---|---|---|
| Collect output | Egg appears at a clear point | Adds an inventory item | Output does not vanish because a day was missed |
| Miss a day | No dramatic negative reaction | Output may wait until care resumes | No injury, death or permanent loss |

## 9.7 Fruit stall and shop economy

The fruit stall is the signature progression system. Selling should require one or two intentional inputs: choose goods or confirm a clearly previewed basket. It then plays a short character/action beat, updates coins and stock, and shows a transaction summary. Do not silently auto-sell everything simply because the player presses E.

Main actions: sell selected goods; fulfil a request; browse stock; buy seeds; buy home/garden decor; upgrade stall. Show item, quantity, unit price and total payout. Protect quest items or show a clear warning before sale. Stall upgrades are visible: new shelf, better awning, fruit display, sign decoration or bigger counter. Upgrades unlock product choices/order capacity rather than hiding an unexplained price multiplier. Reputation comes from varied fulfilled requests and milestones, not repetitive selling of a single cheap item.

## 9.8 Home and decoration

The shared treehouse is the main safe hub. MVP spaces: ground-floor living area, storage and entry; upper-floor kitchen zone and sleeping area. Continuous stairs or a simple interior transition are both acceptable if camera behavior is clear. The player can rest/save, store selected items, display collectibles, review journal and place a small curated set of decorations.

- Storage is distinct from inventory; show counts and never discard items through ambiguous drag/drop.
- Decoration uses curated anchor points or a small snap grid. Physics-based free placement is not required.
- Start with 8–12 items: rug, lamp, plant, wall picture, stool, shelf object, flower pot, basket, keepsake display and kitchen detail.
- Upgrades are visible and expressive, not merely a number. The first may add a shelf or balcony plant instead of a new floor.
- Saving/resting is never locked behind decoration or shop progression.

## 9.9 Day, time, weather and seasons

Recommended starting default is roughly 12 real-time minutes per in-game day, subject to testing. The day/night cycle supports atmosphere and NPC rhythm, not deadline pressure. Players can rest to reach morning. Dialogue and menus pause or suspend time. Crop progress uses game time only, never real-world elapsed time. Rain may water crops automatically if weather exists, but rain is never required to survive. Seasons are stretch/post-launch because each season needs enough art and gameplay differences to justify the effort.

No relationship loss, crop death or permanent missed event can occur solely because a player sleeps or takes a break.

## 9.10 Progression and unlocks

Progression has three parallel tracks: **village restoration**, **stall growth** and **home personalization**. A player should always have at least one meaningful next goal without needing to maximize all tracks.

| Track | Unlock examples | Main reward |
|---|---|---|
| Village restoration | Repaired bridge, reopened gate, improved park | New path, NPC scene, quest and improved ambience |

| Track | Unlock examples | Main reward |
| --- | --- | --- |
| Fruit stall | Extra shelf, order board, special goods | New item choices and a more attractive business |
| Home | Storage, shelf, rug, plants and keepsakes | Self-expression and optional character moments |
| Journal | Fruit, fish, flowers, animals and landmarks discovered | Knowledge, small rewards and completion goals |
| Duo bond | Shared milestones and optional scenes | Dialogue, keepsakes and companion reactions |

# 10. Quest Design and First 30 Minutes

## Quest rules

- Each quest has a plain-language goal, a visible next step and a reward related to the request.
- Use direct verbs: gather, water, feed, find, talk, deliver, decorate, fish and repair.
- Limit early quest chains to 2–3 objectives at a time.
- Completion is determined by authoritative game state, not dialogue text alone.
- Quest items are protected from accidental selling or clearly marked.
- Cozy requests can be paused and resumed. No fail state is used unless a clearly labeled optional challenge is designed later.

## Opening sequence

| Minutes | Experience | Implementation note |
|---|---|---|
| 0–2 | Show forest clearing, Mnki movement and camera | No long cinematic; allow control quickly |
| 2–5 | Swan greets Mnki and points toward home/village | Teach interact, talk and companion presence |
| 5–8 | Gather initial berries and a mango or flower | One clear prompt; immediate inventory feedback |
| 8–11 | Arrive at the shared treehouse; inspect storage/resting spot | Teach home without a long tour |
| 11–15 | Meet Thronn and gently care for one animal | Teach safe care with clear feedback |
| 15–20 | Visit market, speak to grocer, sell a small basket | Show transaction total and coin change |
| 20–25 | Accept first resident request | Journal marks exact item and quantity |
| 25–30 | Complete request; earn decor/seed or stall reward | Show a visible world response, then open free exploration |

## Sample quest: “A Basket for the Village”

1. Swan notices that the market feels quiet and suggests making a basket for neighbors. Objective: talk to the market grocer.
2. Gather 3 berries from the Berry Maze and 1 mango from the Tropical Orchard. Items collected before accepting the quest can count if they are in inventory.
3. Visit the animal farm and gently care for one animal with Thronn.
4. Return to the stall, choose “Prepare basket” and confirm. Only the required items are consumed; unrelated items remain in inventory.
5. Deliver the basket to the Elder in the village park. A short group reaction and a visible decoration/cleanup improvement occur.
6. Reward: coins, a flower seed packet or keepsake, a journal entry and a new optional request.

## Quest data fields

| Field | Purpose |
|---|---|
| `quest_id` | Stable ID, e.g. `q_village_basket_01`; never rely on translated title text |
| Title/description | Localized player-facing text, concise and specific |
| Prerequisites | Quest flags, story milestones, unlocked regions |
| Objectives | Typed tasks: talk, gather quantity, interact, deliver, care, restore |
| Completion condition | Explicit check that all required objectives are complete |
| Rewards | Coins, item IDs, unlock IDs, relationship flag, visual world change |
| Availability | Main, optional, once, repeatable |
| Failure/abandon behavior | Normally no failure; pause and resume |
| Journal summary | Narrative reason and current next action |

## Dialogue samples

**Swan:** “The market feels a little sleepy today. Shall we bring something fresh for the neighbors?”

**Player choices:** “Let’s make a basket.” / “I’ll look around first.”

**Swan:** “Of course. There is no rush. I’ll meet you near the stall.”

**Thronn:** “No need to rush. Hold the food low and let the chicken come to you.”

**Interaction prompt:** “Gently pet the chicken.”

**Success:** the chicken bobs its head; a soft cluck plays; the journal marks Animal Care discovered.

# 11. Items, Inventory and Economy

## Inventory behavior

Use a stack-based grid or list. Every item has a stable ID, display name, icon, category, stack limit, sell value, description and tags such as `quest_protected`, `plantable`, `food`, `gift` or `decor`. Production logic must not identify items by visible display string; labels may change through localization.

- Inventory should be generous early so players learn the world rather than repeatedly emptying bags.
- Common resources stack; decorations and tools are unique entries.
- Transaction preview shows values before selling.
- Quest requirements reserve or pin needed items; selling one warns the player.
- If inventory is full, the world item remains available and the HUD explains why pickup failed.
- Storage in the treehouse is distinct from carried inventory.

## Initial economy values — tuning hypotheses

These are initial test values, not final balance. Adjust after observing earning rate, session length, purchase choices and the time needed to unlock the first stall upgrade.

| Item/activity | Suggested sell value | Source/use | Balance note |
|---|---|---|---|
| Berry | 2 coins | Berry Maze; order ingredient | Common, low value, good for tutorial |
| Mango | 5 coins | Tropical Orchard and stall | Higher value; source regrows |
| Banana | 4 coins | Tropical Orchard | Mid-value category; add after core flow works |
| Coconut | 6 coins | Tropical Orchard | Higher-value visible gathering point |
| Flower | 2 coins | Flower Garden | Sellable or reserved for decor/quests |
| Common fish | 7 coins | Waterfall/ponds | Reward for focused activity |
| Egg | 3 coins | Chicken care | Daily output; must not dominate economy |
| Rice/crop bundle | 3–5 coins | Paddy Field | Tune by grow time and effort |
| Basic seed packet | 3–5 coins purchase | Grocer or quest | Tutorial seeds should not be blocked by money |
| Starter home decor | 8–15 coins purchase | Decor stall or quest | One item should be reachable in a short session |
| First stall upgrade | 40–60 coins | Shelf/canopy/display | Mid-session milestone, not a grind wall |

## Economic safeguards

- Track every coin source and sink in a balance sheet.
- Avoid one item becoming a dominant income exploit unless its rarity is a deliberate, understandable design.

- A quest turn-in, sell action or upgrade cannot trigger multiple times from one input.
- If shop reputation is separate from coins, explain its role clearly: unlock special orders and display options, not hidden price multipliers.
- Do not add premium currency, loot boxes, paid energy, time-skip purchases or randomized paid rewards to the MVP.

## Example basket transaction

A basket contains 3 berries and 2 mangoes. At initial test prices: `(3 × 2) + (2 × 5) = 16 coins`. The shop preview shows each item, quantity, unit value and total. After confirmation, the transaction removes exactly those quantities, adds 16 coins once, increments relevant quest/order counters and plays one sale response. If any required item quantity changed between preview and confirm, validate again and recalculate instead of completing a partial or duplicated transaction.

## Money sinks

The first currency sinks are seeds, home decorations, stall upgrades and selected community repair materials. Keep enough optional sinks that players can choose what to buy. Do not force a player to buy decor to progress a main quest. If a main upgrade is required, award enough coins or materials through a predictable route.

# 12. Interface, Feedback and Accessibility

## HUD philosophy

The world and characters take priority. The exploration HUD shows only the basics: day/time indicator, coin count, one current objective reminder, and a contextual prompt when a valid target is close. Inventory detail belongs in a separate screen. Avoid putting permanent lists of every quest along screen edges.

- **Day/time:** small icon with clear label; no huge countdown clock.
- **Coins:** visible near inventory shortcut; number changes subtly.
- **Context prompt:** bottom-center or near the interaction point; “E — Gather berries” rather than “Interact”.
- **Quest reminder:** one main objective, expandable to the journal.
- **Status messages:** brief, readable and placed where they do not cover subtitles or faces.
- **Menus:** pause single-player simulation or clearly communicate if time continues. Default recommendation is to pause.

## Required UI screens

| Screen | Contents | Rules |
|---|---|---|
| Title menu | Continue, New Game, Settings, Credits, Quit | Confirm overwriting a saved game |
| Inventory | Stacks, categories, descriptions, selected item actions | Show values and quest protection; no accidental destruction |
| Shop | Sell selected goods, orders, buy list, upgrade preview | Show unit values and totals before confirmation |
| Journal | Main quest, side requests, checklist, discovery notes | Explain the exact next action; allow untracking |
| Map | Known regions, player, home, market and discovered quest marker | Do not reveal secrets before discovery |
| Home decoration | Room, furniture choice, snap points, confirm/cancel | Show invalid placement; undo last placement |
| Settings | Audio, controls, graphics, camera, accessibility and language | Immediate feedback and reset defaults |
| Dialogue | NPC name/portrait, text, response options | Skippable after first view; few options at once |
| Pause | Resume, save, settings, return to title | Honest save status; warn only about meaningful unsaved changes |

## Visual hierarchy

Primary actions have the strongest contrast. Secondary actions are quieter. Unavailable actions are disabled with a reason. Quest completion is satisfying but brief. Color is never the only way to communicate a state; pair it with labels, icons or shapes.

## Accessibility requirements

- Remappable keyboard controls for all essential actions.
- Camera sensitivity, invert-Y, smoothing and camera shake options.
- Subtitles for dialogue and meaningful audio cues; text size adjustment.

- Separate volume sliders for music, ambience, effects and dialogue.
- Optional hold/toggle settings for run and repeated actions.
- No mandatory rapid tapping. Fishing’s timing window can be widened or set to an easier mode.
- Icon/pattern support for item rarity and crop states; do not rely on color alone.
- Reduced motion option for camera and interface effects.
- Windowed, borderless and fullscreen support with common resolutions.
- UI focus must remain visible when navigating with keyboard; future controller support must not require mouse-only interaction.

# 13. Visual Direction and Asset Production

## Art thesis

The world should resemble a carefully built miniature set rather than a realistic forest. Oversized roots, rounded leaf masses, curved paths, tree homes, handmade-looking roofs and fruit displays create a sense of scale. Detail is concentrated at player eye level and around landmarks/interactables. Materials are stylized and tactile, with gentle color harmonies and soft light.

## Shape language

| Subject | Shape language | Avoid |
|---|---|---|
| Mnki and Swan | Rounded proportions, clean silhouettes, readable eyes and gestures | Realistic anatomy, cluttered accessories or altered canonical outfits |
| Tree Village buildings | Organic stacked volumes, visible beams, curved roofs and simple trim | Photorealistic noise or generic modern boxes |
| Nature | Clustered leaves, color groups of flowers, simplified bark and layered foliage | Thousands of unique tiny leaves that harm performance/readability |
| Water | Clear stylized surface, restrained highlights and gentle movement | Noisy reflections that obscure fishing targets |
| Collectibles | Strong shape/color contrast at normal camera distance | Collectibles visually identical to decoration |
| UI | Soft panels, cream backgrounds, forest-green text and sparing coral/gold accent | Heavy frames that cover the world |

## Palette proposal

| Color | Use | Reference |
|---|---|---|
| Forest green | Main environment, headings, UI anchor | #173B31 |
| Leaf sage | Foliage highlights, secondary panels | #DDE9D7 |
| Warm cream | Interior lighting, menus and UI surfaces | #FBF7ED |
| Fruit coral | Market awnings and accent color | #E77F6E |
| Water teal | Ponds, fishing feedback and water | #78C9D4 |
| Harvest gold | Coins, highlights and warm sunlight | #E4B96B |
| Wood brown | Treehouse, stall, trunks and baskets | #8B6749 |

## Scale and camera consistency

Adopt a world-unit convention before importing final models: one Godot unit can be treated approximately as one meter, with stylized character height around 1.4–1.7 units as a starting point. Verify the chosen character scale in the actual camera view. The player should remain identifiable at thumbnail scale. Use consistent pivots: character at ground contact, doors at hinge, props at placement base, trees at trunk base. Apply transforms and create collision proxies rather than using every render polygon as collision.

Large trees and buildings may use LODs or simpler distant representations if profiling requires them. Do not add impostors or complexity by default; measure first. Materials should have a coherent roughness range so wood, leaf, stone, water and cloth remain visually distinct without photorealism.

## Asset production pipeline

1. Block out paths and spaces with Godot primitives and gray materials. Validate camera, player scale and interaction distances.
2. Approve the character turnarounds, palette, material language and prop scale chart.
3. Create and rig Mnki first; verify headband, hoodie, pants, shoes and tail in-engine.
4. Create and rig Swan with explicit wing geometry and rig checks; her hood/beak/crown silhouette must remain readable.
5. Build modular environment kits: roots/trunks, canopy clusters, path edges, fences, roofs, beams, doors, stalls, crates, flowers, plants, stones and interior modules.
6. Add collision proxies and LODs where measured. Import glTF/GLB assets and verify scale, pivots, materials, animation tracks and origin.
7. Review every asset in-engine for camera distance, clipping, readability, lighting and performance before marking it approved.

## Minimum animation set

| Animation | Mnki | Swan | Priority |
|---|---|---|---|
| Idle/turn in place | Required | Required | MVP |
| Walk/run | Required | Required for companion | MVP |
| Start/stop/turn blends | Required | Desired | Quality bar |
| Jump/land | Required | Optional initially | Stretch for Swan |
| Gather/pick fruit | Required | Short reaction | MVP |
| Pet/feed animal | Required | Reaction/assist | MVP |
| Fish/cast/catch | Required if rod used | Watch/celebrate reaction | MVP |
| Shop/deliver basket | Required | Shared celebration/comment | MVP |
| Happy/curious/thoughtful reactions | Required | Required | MVP |
| Sleep/wake | Required or transition | Shared scene support | MVP |
| Swan flight | Not applicable | Not required | Post-MVP, avoid scope creep |

## Character QA checklist

- The gold curled headband is visible from expected angles and does not float or intersect the hair.
- The curled tail is always attached, does not vanish behind the legs for long stretches, and does not collide erratically with the environment.
- Swan’s large white wings are present, correctly attached and do not clip badly into the torso during walk/idle.
- Swan’s hood, orange beak and small gold crown remain readable at gameplay scale.
- Pink pleated skirt and white top maintain their canonical colors; shoes and socks remain present.
- No animation causes characters to pass through a counter, animal, companion or camera blocker.

# 14. Audio Direction

Audio makes the miniature world feel alive without being busy. Use several clear location layers and short action sounds. Music should feel warm, lightly whimsical and spacious so ambience and dialogue still read. Do not copy another game’s recognizable musical or sound signature.

## Music areas

| Cue | Feeling / instrumentation | Use |
|---|---|---|
| Main theme | Gentle plucked strings, soft piano, light woodwind or warm synth | Title menu and thematic returns |
| Sunshine Forest | Airy, open, slow pulse | Opening and exploration |
| Village Market | Light rhythmic taps and warm melody | Market; duck under shop dialogue |
| Farm/orchard | Organic percussion and subtle folk texture | Gather/cultivation activity |
| Treehouse | Intimate, mellow, familiar | Home, rest and shared scenes |
| Milestone | Small rise and resolved warm chord | Quest completion, stall upgrade, path restored |
| Evening | Slower and lower density | Wind-down and day transition |

## Sound effect rules

- Distinct cues for gathering, harvest, fishing bite, catch, petting, feeding, sale, upgrade, quest progress and save success.
- Every critical audio cue has a visual backup for players at low volume or without audio.
- Duck ambience under dialogue and tutorial guidance.
- Keep prominent bird calls from repeating every few seconds; loop points should be seamless.
- Provide separate music, ambience, effects and dialogue volume controls.
- Use original, licensed or clearly permitted audio and keep source/license records in the asset register.

# 15. Godot 4.7.2 Architecture

## Engine target and render policy

Target **Godot 4.7.2 stable**, released 18 August 2026 in the official release archive [1]. Pin the project to this version for reproducible builds and version-control the project before engine upgrades. Verify exact properties and API details in the installed 4.7.2 editor and official documentation; this GDD specifies architecture, not a replacement for editor testing.

Use Forward+ as the initial Windows-PC renderer for stylized lighting, shadows and environment effects. Keep art direction compatible with a lower-feature fallback if broader hardware support becomes necessary. Do not make high-end post-processing essential to gameplay readability. Profile shadow distance, transparency, foliage overdraw and water before adding more effects.

## Recommended node/system choices

| Need | Recommended approach | Implementation note |
|---|---|---|
| Player movement | `CharacterBody3D`, `CollisionShape3D`, camera pivot and `Camera3D` | Designed for scripted character movement and collision handling [3] |
| NPC/companion navigation | `NavigationRegion3D` with baked navigation mesh; `NavigationAgent3D` or waypoints | Use nav only when the mesh is ready; recover from unreachable targets [4] |
| Interactables | `Area3D`, spatial query or scanner plus shared interaction interface | Consistent behavior for plants, NPCs, shop and animals |
| Data definitions | Custom `Resource` classes saved as `.tres` | Editor-friendly; stable IDs separate from display text |
| World modules | Reusable `PackedScene` scenes | Modular region/building/prop ownership |
| Cross-system communication | Typed signals for inventory, quests, coins, day changes and unlocks | Reduces direct coupling |
| Persistence | Versioned save schema under `user://`, with validation and backup | Use tested save/load design, not arbitrary scene-tree dumps [5] |
| UI | Reusable `Control` scenes per major screen | UI issues commands; systems remain authoritative |
| Localization | Translation resources/CSV and keys | Do not hardcode display text into gameplay logic |

## Input Map

| Action ID | Default | Notes |
|---|---|---|
| `move_forward` / `move_back` | W / S | Camera-relative movement |
| `move_left` / `move_right` | A / D | Camera-relative movement |
| `look` | Mouse motion | Captured only while gameplay owns input |
| `interact` | E | Shared world interaction |
| `jump` | Space | Optional; not required for core progress |
| `run` | Left Shift | No stamina requirement |

| Action ID | Default | Notes |
|---|---|---|
| inventory | Tab / I | Final default after playtest |
| journal | J | Current quest/discoveries |
| map | M | Region overview |
| pause | Esc | Pause and free mouse |
| cancel | Esc / right mouse | Context-sensitive |

## Production project layout

```
res://
 project.godot
 scenes/
   boot/boot.tscn
   menus/title_menu.tscn
   world/tree_village_world.tscn
   regions/sunshine_forest.tscn
   regions/waterfall_ponds.tscn
   regions/berry_maze.tscn
   regions/flower_garden.tscn
   regions/paddy_farm.tscn
   regions/tropical_orchard.tscn
   regions/animal_farm.tscn
   regions/guardian_gate.tscn
   regions/village_park.tscn
   interiors/shared_treehouse.tscn
   regions/colony_one.tscn
   regions/colony_two.tscn
   regions/market_square.tscn
   characters/mnki.tscn
   characters/swan_companion.tscn
   characters/thronn.tscn
   animals/chicken.tscn, sheep.tscn, cow.tscn
   interactables/gather_node.tscn, farm_plot.tscn
   interactables/fishing_spot.tscn, shop_counter.tscn
   ui/hud.tscn, inventory.tscn, shop.tscn, journal.tscn, map.tscn
 scripts/
   core/game_state.gd, save_manager.gd, time_manager.gd
   core/scene_router.gd, audio_manager.gd
   player/mnki_controller.gd, camera_rig.gd
   player/interaction_scanner.gd
   companion/swan_companion_ai.gd
   systems/inventory_system.gd, economy_system.gd
   systems/quest_system.gd, farming_system.gd, fishing_system.gd
   systems/animal_care_system.gd, day_cycle_system.gd
   systems/dialogue_system.gd, shop_system.gd
   systems/world_state_system.gd
 data/items/, crops/, fish/, animals/, quests/, npcs/, shop_upgrades/
 art/characters/, environments/, props/, materials/, vfx/
 audio/music/, ambience/, sfx/, dialogue/
 localization/, tests/, docs/
```

Keep global managers few and deliberate. Suggested autoloads: GameState, SaveManager, TimeManager, AudioManager and SceneRouter if needed. Inventory, Quest, Shop and Farming can be services owned by GameState or the world scene. Do not make every system global for convenience. Every system needs one clear source of truth and typed signals for state changes.

## Example world scene tree

```
TreeVillageWorld (Node3D)
 WorldEnvironment
 DirectionalLight3D / optional fill lights
 NavigationRegion3D
 GroundAndPaths
 Landmarks
 RegionModules
 Interactables
 NPCs
 Animals
 PlayerSpawn
 PlayerRoot (Mnki)
 CompanionRoot (Swan)
 WorldStateMarkers
 UIHost (CanvasLayer)
```

Avoid putting the whole game into one large `main.gd`. The current generated scene is useful for fast prototyping, but production logic must be broken into reusable systems, scenes and data resources.

# 16. Data, Signals, AI and Save/Load

## Data-driven content

Content creators should be able to add an item, crop, quest or shop offer without editing many unrelated scripts. Stable IDs are machine-readable and never depend on visible/localized titles. If an ID is renamed or removed, a save migration or compatibility fallback is required.

### ItemDefinition contract

```
ItemDefinition (Resource)
 id: StringName              # e.g. item_mango
 display_name_key: StringName   # localization key
 description_key: StringName
 category: ItemCategory
 icon: Texture2D
 stack_limit: int
 base_sell_value: int
 buy_value: int
 tags: Array[StringName]
 can_sell: bool
 quest_protected_default: bool
```

### CropDefinition contract

```
CropDefinition (Resource)
 id: StringName
 seed_item_id: StringName
 harvest_item_id: StringName
 growth_stages: Array[Texture2D or PackedScene]
 game_days_to_maturity: int
 water_required_per_day: bool
 yield_min: int
 yield_max: int
 regrow_days: int
 season_tags: Array[StringName]  # future-ready; not MVP-critical
```

### Quest objective types

| Type | Parameters | Completion event |
|---|---|---|
| TalkTo | NPC ID | Dialogue node marked complete |
| CollectItem | Item ID, quantity | Inventory quantity change; design decides if previously owned items count |
| InteractWith | Target ID, count | Interaction completed signal |
| DeliverItems | Target NPC, item list | Validated delivery transaction |
| CareForAnimal | Animal/species, action | Care action confirmed |
| SellValue | Minimum total and optional filter | Shop transaction completed; prevent duplicate payout |
| DiscoverRegion | Region ID | Discovery flag persisted |
| RestoreWorldObject | World-object ID | Visual state and unlock flag updated |

## Signals — source of truth

| Signal | Emitted by | Consumers |
|---|---|---|
| `inventory_changed(item_id, delta)` | InventorySystem | HUD, inventory, quest tracker, save-dirty flag |
| `coins_changed(old_value, new_value)` | EconomySystem | HUD, shop UI, feedback audio, save system |
| `quest_objective_updated(quest_id, objective_id)` | QuestSystem | Journal, HUD, dialogue eligibility |
| `quest_completed(quest_id)` | QuestSystem | Reward handler, NPC dialogue, world state, audio |
| `day_started(day_index)` | TimeManager | NPC schedules, crops, animal output, lighting |
| `item_gathered(item_id, quantity, source_id)` | Gather/Inventory system | Quest tracking, tutorial, approved analytics |
| `region_unlocked(region_id)` | WorldStateSystem | Map, gate visuals, quest availability |
| `shop_transaction_completed(total, items)` | ShopSystem | HUD, quest tracking, save state and sound |

Inventory owns quantities. Shop owns offers and transaction validation. Economy owns the coin balance and price calculation. QuestSystem reads events and flags but does not directly edit crop meshes or inventory internals. WorldStateSystem owns unlock flags and restoration IDs. UI reads system data and sends commands; it is never the authoritative source of currency, inventory or quest completion.

## Swan companion state machine

| State | Entry condition | Behavior | Exit condition |
|---|---|---|---|
| Follow | Normal exploration and path clear | Follow a point behind/to the side of Mnki; update target at a modest interval | Player interacts or reaches a special point |
| Wait | Player fishes, shops, talks or uses a small interaction zone | Move to a non-blocking wait point and play subtle idle | Interaction ends or player moves away |
| React | Discovery, milestone or selected event | Play one short animation or line without taking control | Reaction finishes |
| NavigateToPoint | Quest scene, home or scripted meeting | Use nav mesh or waypoints; validate destination | Arrive or recovery timeout |
| Recover | Path invalid or companion too far away | Try a safe point; only rejoin behind the player and preferably out of view | Safe position reached |

Recompute follow points at a controlled interval rather than blindly every rendered frame. Do not target the exact player position, which can cause overlap and jitter. Log recovery events during development so bad navigation is not hidden by recovery logic. Swan never blocks a mandatory doorway or quest target.

## Saving and loading

A cozy game encourages long-term return, so lost progression is a critical defect. Use an explicit, versioned save schema. Do not dump the entire live scene tree or depend on fragile node paths.

**Minimum saved fields:** schema version and timestamp; day/time; current safe location; coins; inventory and storage; quest states/objectives; unlocked regions; world restoration flags; NPC conversation flags; crop plot IDs/stages/watering/growth day; animal care states/output flags; stall upgrade/reputation/orders; home decoration IDs/positions; journal discoveries; and settings if stored in the same file.

**Save moments:** manual save from pause/home; autosave after important milestones such as quest completion, shop upgrade and sleeping. Never save during a partial transaction. Complete the state mutation first, then commit.

**Safety requirements:** write to a temporary file, confirm success, then replace the primary file; keep a backup where feasible. Validate unknown IDs and schema versions on load. Unknown content should be skipped/recovered safely, not crash the game. Show an honest saved indicator. Do not claim Steam Cloud support before implementation and testing.

Example schema (illustrative only):

```
{
 "schema_version": 1,
 "game_day": 3,
 "coins": 42,
 "inventory": {"item_berry": 4, "item_mango": 2},
 "completed_quests": ["q_village_basket_01"],
 "unlocked_regions": ["sunshine_forest", "market_square"],
 "shop": {"upgrade_level": 1, "reputation": 2},
 "world_flags": ["bridge_path_restored"],
 "home_decor": []
}
```

Test new save, overwrite, reload, missing file, truncated/corrupt file, old schema, invalid IDs, interrupted save and repeated save/load during crop growth. Settings should be recoverable without resetting game progress.

# 17. Level Design and Environmental Storytelling

## Landmark hierarchy

Each region has a primary landmark visible from the entrance, secondary markers every short stretch and micro-details that reward slowing down. The red canopy identifies the market; the waterfall marks the pond route; three guardian statues mark the village threshold; the shared treehouse is the home silhouette. Landmark silhouettes should remain readable through different lighting states.

## Path design

- Main routes are wide, low-friction and clear of collision clutter. Exploratory branches may be narrower but must remain camera-friendly.
- Curves reveal destinations in stages rather than hiding all landmarks behind dense foliage.
- Contrast paths and ground cover so players naturally understand where walking is expected.
- Place interaction points with character and companion clearance; never put key objects in doorways, narrow bridges or steep slopes.
- If a region is blocked until a quest, make the gate visible and provide an explanation.
- Keep the first thirty meters visually simple enough for onboarding.
- Use visual barriers (roots, fences, ponds or terrain) instead of invisible walls wherever possible.

## Environmental story details

| Detail | Implied story | Possible action |
|---|---|---|
| Half-repaired bridge | The community once maintained shared paths | Deliver materials or complete a helper quest to finish it |
| Empty market crate | The stall needs produce from nearby regions | Fill display crates when goods are sold |
| Different offerings at the statues | The community remembers Land, Water and Forest | Read plaque or make a flower/water offering during a quest |
| Thronn’s animal-care board | Animals are part of community routine | See a daily optional care request |
| Keepsake near a treehouse window | Mnki and Swan have personal history | Inspect for optional dialogue or journal entry |
| New park flower bed | Player contribution remains in the world | Flowers persist after restoration quest |

## Region acceptance checklist

Each region must have: a unique landmark, one distinct dominant color note, a readable route back to the village, one clear activity or reason to visit, a short audio identity, appropriate collision, stable save IDs for persistent state, and at least one camera test at the entrance and the most cramped point. Decoration must not obscure important interactables or companion navigation.

# 18. QA and Acceptance Criteria

## Test matrix

| Area | Essential test cases | Severity if broken |
|---|---|---|
| Movement/camera | Walls, bridge, slopes, stairs, jump, camera collision, run transitions, menu mouse release | High if stuck, falling through or loss of control |
| Interaction | Two close targets, removed target, out-of-range target, occlusion, key press during dialogue, repeated input | High if wrong action or duplicate transaction |
| Inventory/economy | Full inventory, protected item, zero-value sale, quantity/coin balance, duplicate input, negative quantity | Critical if items/currency lost or duplicated |
| Farming | Correct/invalid seed, watering, day advance, mature crop, double harvest, save mid-growth | High if growth softlocks or duplicates |
| Fishing | Bite timing, no bite, cancel, companion nearby, full inventory, save/load | Medium-high if softlock/repeated reward |
| Animals | Pet/feed, wrong food, repeated input, next-day output, persisted care state | Medium-high if care/output breaks |
| Quests | Locked prerequisite, pre-owned item, partial objective, repeated turn-in, missing NPC, save midquest | Critical if quest softlocks or pays twice |
| Companion | Blocked path, inaccessible target, house transition, distance recovery, navmesh update | High if Swan blocks progression |
| Saving | New/overwrite/reload, corrupt file, schema migration, backup recovery, close during save | Critical if progress lost |
| Performance | Dense foliage, water, shadows, NPC cluster, scene transitions and long session | High if severe stutter/crash |
| UI/accessibility | Key remap, subtitle size, audio sliders, reduced motion, fullscreen, focus navigation | High if essential setting broken |

## Definition of Done for every system

- Works from new game to save/load without developer-only inspector edits.
- Gives appropriate prompt, animation/state change and HUD/audio feedback.
- No duplicate rewards, impossible state, null-reference spam or fragile hard-coded path dependencies.
- Content is data-driven where appropriate and uses stable IDs.
- Includes a successful test, invalid-input test, edge-case test and persistence test.
- A teammate can use the feature from a short written instruction.

## Performance target

Target 60 FPS at 1920 × 1080 on a reference mid-range Windows PC, whose exact CPU/GPU/RAM baseline must be recorded before alpha. Measure frame-time distribution in the market hub, foliage-heavy forest, pond, NPC group and interior. If performance is poor, reduce transparent foliage overdraw and shadow range before compromising character readability. Profile on real hardware; editor performance alone is not enough.

## Global player-experience acceptance

- A new player understands movement, interaction and the first quest without coaching.
- Player can gather two resources, speak to Swan/Thronn, care for an animal, sell goods, spend earnings and finish a quest.
- A completed quest produces a visible world or character response.
- Stopping and resuming later does not lose meaningful progress.
- No mandatory action forces camera clipping, blocks movement or leaves the player stuck.

# 19. MVP Scope and Production Roadmap

## Required MVP

| Feature | Required result | Exit gate |
|---|---|---|
| Mnki controller | Stable third-person walk/run/jump, camera collision and remapping | 20-minute test without control blockers |
| Swan companion | Follow/wait/react states, contextual comments and safe recovery | Never blocks critical path or visibly teleports repeatedly |
| World | Compact Tree Village with all key landmarks; colonies may be connected compact subdistricts | New player can navigate home, market, forest, pond and farm without a guide |
| Gathering/inventory | Berry, mango, flower, fish and crop/egg item paths with stack rules | No loss/duplication; full inventory is clear |
| Farming | Multiple crops, visible stages, watering and harvest | Crop state survives sleep and save/load |
| Fishing | One satisfying cast/bite/catch loop | Clear feedback, no softlocks |
| Animal care | Cow, sheep and chicken; pet/feed/groom loop | Care action and day state persist |
| Fruit stall | Sell transaction, prices, orders and one visible upgrade | Transaction summary and coin persistence work |
| NPCs/quests | Thronn, elder, gardeners, farmer/fisher/grocer roles and short chain | Main demo quest completes end-to-end |
| Home | Treehouse, rest/save, storage and small decoration set | Clear function and visible personalization |
| Polish | Soundscape, UI, onboarding, accessibility basics and save/load safety | External testers complete demo without live coaching |

## Stretch goals — only after core stability

- Controlled switching between Mnki and Swan with reliable interaction parity.
- More animal outputs such as milk or wool.
- Cooking recipes or a deeper crafting system.
- Additional weather states and seasonal crops.
- More home rooms and freeform decoration.
- Additional shop types, second upgrade tier and more orders.
- A larger Three Guardians story arc.

## Post-launch ideas, not first-release promises

- Enlarged second residential district and further woodland areas.
- Additional story chapters and character keepsakes.
- Optional outfits that preserve canonical recognition.
- Photo mode, expanded collectible sets and advanced home customization.
- Controller-first polish/Steam Deck compatibility if not completed at launch.
- New languages after audience signals justify the expense.

## Explicit non-goals

No online multiplayer or trading; no competitive leaderboard; no combat-first enemies; no permanent crop death or animal death; no procedural world generation; no paid energy, loot boxes, time-skip purchases or premium currency; no visual or narrative direction that replaces Mnki and Swan as the product's core identity.

## Production roadmap

Planning assumption: an experienced solo developer owns game design and code, with asset help as available. A focused full-time path may take about nine months; part-time work or fully custom assets/animation may extend the project to 12–18 months or more. Dates are hypotheses, not guaranteed deadlines.

| Phase | Approx. duration | Work | Gate / deliverable |
|---|---|---|---|
| P0 — Scope lock | 1–2 weeks | Canon sheets, greybox map, hardware baseline, core loop and cut list | Approved brief and checklist |
| P1 — Controller slice | 3–4 weeks | Mnki temporary model, movement, camera, collision, interaction scanner and one region | Movement/gathering reliable |
| P2 — Core loop | 4–6 weeks | Inventory, gathering, first crop, stall sell transaction, coins and HUD | Gather → sell → spend works safely |
| P3 — Duo and quest slice | 3–5 weeks | Swan follow/wait/react, Thronn, quest system and opening story beats | First 20-minute quest completes |
| P4 — Activity systems | 5–7 weeks | Fishing, animal care, crops, storage, day cycle and stall upgrade | Three core activities persist correctly |
| P5 — World production | 6–8 weeks | Compact regions, treehouse, landmarks, NPC routines and audio | Content complete for alpha path |
| P6 — Alpha/QA | 4–6 weeks | Save/load, UX, accessibility, profiling and external tests | No critical/high blockers |
| P7 — Beta/launch prep | 4–6 weeks | Store assets, trailer, demo, build pipeline, credits and localization pass | Release candidate validated on target PCs |

## Milestone rules

- Do not start another major system while a critical save, interaction, movement or quest defect remains.
- Lock new features in the final 20% of production unless they fix a release blocker.
- At each phase exit record build number, known issues, performance result and next dependencies.
- If the schedule slips, reduce content volume or art variation before cutting save safety, onboarding, controller feel or feedback.
- Treat player confusion as evidence that the design needs improvement, not as a tester's failure.

# 20. Budget, Pricing and Revenue Scenarios

## Budget framework

These are planning ranges in Indian rupees, not supplier quotes. They exclude salary/opportunity cost for the developer, income tax, larger custom-content overruns and future live support. Godot has no engine royalty, but third-party assets, audio and services can still cost money and carry license limits.

| Spend category | Lean solo / asset-assisted | Small outsourced support | Notes |
|---|---|---|---|
| Characters, rig and animation | ₹25,000–₹80,000 | ₹1,00,000–₹2,50,000 | Hero quality is highly visible; reserve budget here |
| Environment kits and props | ₹15,000–₹50,000 | ₹1,00,000–₹2,00,000 | Use modular kits and custom hero landmarks; avoid conflicting kits |
| Audio, music, licenses | ₹5,000–₹25,000 | ₹20,000–₹60,000 | Record source and license terms |
| QA and playtesting | ₹10,000–₹30,000 | ₹40,000–₹1,00,000 | Include new-player and save/load testing |
| Store art, trailer, marketing | ₹15,000–₹50,000 | ₹50,000–₹1,50,000 | Capture genuine gameplay; avoid misleading mockups |
| Tools and contingency | ₹15,000–₹50,000 | ₹40,000–₹1,00,000 | Includes unplanned asset and compatibility work |
| **Indicative total cash spend** | **₹85,000–₹2,85,000** | **₹3,50,000–₹8,60,000** | Excludes developer labor, taxes and major overruns |

## Budget discipline

- Initial prototype spending cap: target ₹50,000 or less before the core loop passes external playtesting.
- Do not commission many final models until the hero rig, camera scale and art direction work in-engine.
- Check every asset’s commercial-use terms, modification rights, redistribution restrictions and source-file availability.
- Track monthly burn: cash spent, hours invested, remaining scope and milestone confidence.
- Include Steam publishing costs and applicable taxes in the release plan. The current Steam Direct documentation describes a US$100 fee per product and recoupment after US$1,000 in adjusted gross revenue, subject to Steam’s terms [6]. Re-check at time of publication.

## Staffing ownership

| Role | Owns | Part-time possible? |
|---|---|---|
| Designer/producer | Scope, pacing, economy, quests, milestone gates, playtest decisions | Must be actively owned throughout |
| Gameplay programmer | Controller, inventory, shop, farming, fishing, NPC/companion, save/load | Possible, but continuity helps |
| Character artist/rigger | Canonical Mnki/Swan models, rigs and core animation | Yes, against a locked brief and review loop |

| Role | Owns | Part-time possible? |
|---|---|---|
| Environment artist | Modular kit, landmarks, foliage, props and level dressing | Yes, in milestone packages |
| Audio designer | Ambience, action cues, music mix and settings | Yes |
| QA tester | New-player tests, regressions, save/load, performance and edge cases | Yes; intensify near alpha |

## Pricing hypothesis

Recommended model: premium one-time purchase on Steam. For a modest first release, test a broad international list-price hypothesis around **US$9.99–US$12.99**. A larger, deeply polished game may justify more. A possible India price hypothesis is **₹499–₹699**, but check current Steam regional pricing recommendations and comparable releases before fixing it. These are starting hypotheses, not a statement about current competitor prices.

Do not set price because another game uses it. Match price to delivered content, polish, audience feedback, demo response and regional purchasing power. The complete core loop should be included at purchase. Later paid expansions may add substantial areas/story, not remove expected basic functions.

## Revenue scenarios — illustrative, not forecasts

Assumption for simple comparison: ₹600 average customer spend per copy and 70% illustrative developer proceeds after platform share, before additional taxes, refunds adjustments, marketing, production and developer time. Actual payout varies with discounts, regional pricing, refunds, taxes and platform terms.

| Copies sold | Illustrative gross customer sales | Illustrative 70% proceeds | Interpretation |
|---|---|---|---|
| 1,000 | ₹6,00,000 | ₹4,20,000 | Small launch; may not recover production cash plus time |
| 5,000 | ₹30,00,000 | ₹21,00,000 | Meaningful traction if cash costs stay controlled |
| 20,000 | ₹1,20,00,000 | ₹84,00,000 | Strong indie result; not a baseline expectation |
| 50,000 | ₹3,00,00,000 | ₹2,10,00,000 | Breakout scenario requiring strong discovery and retention |

### Cash break-even examples

| Eligible cash cost | Copies at ₹420 illustrative proceeds/copy |
|---|---|
| ₹2,00,000 | 477 |
| ₹5,00,000 | 1,191 |
| ₹10,00,000 | 2,381 |
| ₹20,00,000 | 4,762 |

This break-even model covers only stated cash cost, not labor value, income tax, future support, cost of funds or overhead. Cash break-even is not the same as a profitable business if the project uses a year of unpaid full-time work.

## Commercial validation gates

- Validate enjoyment and core-loop repetition before increasing the art budget.
- A public demo must represent real game quality. Steam’s official guidance says demos should be a high-quality, well-scoped sample that leaves players interested in the full game [2].
- Make the Coming Soon page public when art direction, screenshots and gameplay promise are stable. Steam says the page can build interest and wishlists; current documentation requires it to be live at least two weeks before release for new products [7].
- Wishlist growth is a signal, not a sales guarantee. Track traffic source, demo feedback and production cost alongside the count.

# 21. Marketing and Launch Plan

## Brand sentence

**“A tiny tree village. Two best friends. A world that grows kinder every day.”**

## Content pillars

| Pillar | Example post/video | Proof provided |
|---|---|---|
| Character warmth | Swan reacts to a decorated room; Mnki celebrates an animal | Shows that characters are expressive and central |
| Miniature world | Walk from waterfall to market in a 15-second camera shot | Shows exploration, scale and visual identity |
| Satisfying loop | Gather mangoes → stock stall → fulfil order → unlock shelf | Shows that the economy is tactile, not only a menu |
| Cozy routine | Water crops in morning, fish in afternoon, return home at sunset | Shows calm daily rhythm |
| Build in public | Before/after environment, a bug fixed or an animation pass | Builds trust and attracts useful feedback |

## Timeline

| Stage | Output | Decision |
|---|---|---|
| Early prototype | Short movement/environment clips and private feedback | Validate that the miniature art direction reads clearly |
| Vertical slice | Gameplay clip with Mnki, Swan, one sale and one quest | Publish a page only when product promise is stable |
| Store preparation | Capsule art, genuine screenshots, feature description, gameplay trailer and FAQ | Every claim must match a shipped or clearly planned feature |
| Demo preparation | 20–40 minute polished sample | Players should complete it without developer coaching; never present a broken placeholder as a commercial demo |
| Launch runway | Updates, creator outreach, community posts and bug notes | Judge audience quality and feedback rather than buying broad low-intent traffic |
| Post-launch | Patch notes, support replies and roadmap review | Prioritize defects and content that align with pillars |

## Steam page checklist

- **Hook:** Introduce Mnki, Swan, Tree Village and their fruit stall in one clear sentence.
- **Features:** List only demo-supported activities: explore, grow, fish, care for animals, run the stall, decorate and help villagers.
- **Screenshots:** Show exploration, farming/fishing, animals, stall UI, the shared home and character moments.
- **Trailer and store facts:** Use genuine gameplay to show the cozy loop and characters; end with the title and call to action. State supported languages, controls, accessibility and PC requirements accurately.

# 22. Risks, Privacy and Production Governance

## Risk register

| Risk | Likelihood/impact | Early warning | Mitigation |
|---|---|---|---|
| Scope creep — every area becomes unique/large | High/High | New assets arrive faster than a playable loop | Lock MVP, reuse kits, monthly scope review |
| Character quality below brand standard | Medium/High | Silhouette/outfit fails review | Canon sheets and in-engine approval before bulk art |
| Swan navigation feels broken | High/Medium | Blocks doors, visible teleports, interrupts interactions | Follow/wait/react states, safe points and regression tests |
| Economy becomes grindy or exploitable | Medium/High | One item dominates income; repeated sale grants coins | Economy spreadsheet and transaction tests |
| Save corruption/lost progress | Medium/Critical | Partial files or reload mismatch | Versioned atomic saves, backup and regression suite |
| Art performance too expensive | Medium/High | Foliage, transparency, water or shadows stutter | Profile early; reduce overdraw and shadow range |
| Marketing starts too late | High/Medium | No audience or store page when content is nearly done | Share stable clips, test demo and prepare page once ready |
| Price/content mismatch | Medium/High | Players report insufficient content for price | Re-scope, improve value or lower price after testing |
| Asset license ambiguity | Low-medium/High | Missing license record or redistribution restrictions | Asset register and pre-integration check |
| Marketing overpromises | Medium/High | Trailer shows absent systems | Tie all claims to scope status and producer approval |

## Privacy and community trust

A single-player game can be developed without invasive tracking. Prefer moderated tests, optional surveys and bug reports. If telemetry is later added, collect only what answers a defined design question, avoid personal data, disclose it clearly and follow applicable privacy and platform rules. Never transmit save files automatically for analytics; ask the player to attach one only when intentionally submitting a support report.

Useful low-risk aggregate events may include tutorial-step completion, failed interaction count and successful shop transaction. Do not collect unrelated browsing data, personal files, chat logs or sensitive information. The game should not use manipulative streaks, fear-of-missing-out rewards or punishment for absence.

## Production governance

- **Version control and quality:** Keep source control and tagged milestones; give every task a testable definition of done. Fix data loss, softlocks and crashes before core-loop issues or cosmetic polish.
- **Asset and scope records:** Track asset ownership, source, license, attribution and modification permissions; log changes after scope lock. Accept features only when they strengthen a pillar, fit budget/schedule, and protect persistence and accessibility.

# 23. Existing Prototype Audit and First Refactor

The supplied prototype is an early code-generated proof of concept. It contains a meadow/path, market counter, home mass, pond, waterfall, bridge, trees, flowers, berry bushes, mango trees, crop plots, basic fishing interaction, cow/sheep/chicken placeholders, Thronn and Swan interactions, a shop action, a day counter and a simple HUD. Much of the scene is generated with primitive meshes, and one main script currently owns many systems. This is useful for fast experimentation, not yet a commercial structure.

| Area | Current prototype state | Required production upgrade |
|---|---|---|
| Characters | Placeholder representations; no final model/animation | Canonical models, rigs, materials, tail/wings and reactions |
| World | Primitive boxes/spheres and simple materials | Modular region scenes, designed paths, final lighting/collision and camera tests |
| Interaction | Nearest-target scan plus metadata logic | Shared interface, focus priority, occlusion policy, typed data and edge-case validation |
| Gathering | Berry/mango increase inventory and disappear | Data-driven sources, regrowth, persistent IDs and visible source states |
| Farming | Plot interaction grants generic crop item | Plant/water/growth/harvest state machine and visual stages |
| Fishing | Randomized catch outcome | Bite cue, cancel/retry, fish table and accessibility tuning |
| Animals | Care action grants output or coins immediately | Species animation, daily state, output rules and persistence |
| Shop | Sells all listed inventory automatically and increments counter | Selection/confirmation, protected quest items, orders and visible upgrade state |
| Quests | Single starter boolean | Data-driven objectives, prerequisites, reward transaction, journal and save/load |
| Day cycle | Short timer increments day display | Time-of-day lighting, sleep transition, NPC schedule and crop/animal timing |
| Save/load | Not implemented in current README snapshot | Versioned save, manual/autosave, backup, migration and tests |
| Assets/polish | Placeholder scene | Art pipeline, audio, accessibility, performance, QA and Steam preparation |

## First refactor sequence

1. Keep a version-controlled copy and tag a known-working baseline.
2. Resolve parse/runtime errors and confirm clean launch in Godot 4.7.2 before architectural changes.
3. Separate player controller, camera and interaction scanner.
4. Create ItemDefinition resources and one InventorySystem/EconomySystem authority.
5. Replace generic target metadata handling with a shared interaction contract.
6. Implement one complete quest and a safe shop transaction backed by save/load.
7. Add Swan follow/wait/react behavior with tests for blocked paths and interaction suppression.
8. Replace placeholder hero meshes only after controller scale and camera framing are stable.

9. Expand the world area by area using reusable modules instead of growing a single script.

## Player controller implementation notes

Separate locomotion, camera and interaction. Movement direction is computed relative to camera yaw. Use explicit `Vector3` types when inference is ambiguous. All physics movement happens in the physics process. Keep body collision separate from the animated visual skeleton. Expose speeds, acceleration, gravity, jump velocity and turn rate as tunable parameters. The camera must not write character world velocity.

## Interaction scanner algorithm

1. Collect candidates from an interaction group or Area3D sensor.
2. Discard freed, disabled, invalid or too-distant targets.
3. Compute direction from camera to target.
4. Apply view-cone and optional line-of-sight checks.
5. Rank by view alignment, distance and explicit priority.
6. Display one prompt.
7. Revalidate target state when E is pressed.
8. Call the interaction once.
9. Update HUD from system signals.

## Day and companion optimization

Use a central clock in in-game minutes. Crops and NPCs react to day/time events instead of polling every frame. Pause/gate time during dialogue and menus according to settings. Swan recomputes its follow target at a controlled interval, waits during player interactions and uses safe recovery only when meaningfully far away. Debug logs should expose every recovery during development.

# 24. Appendices and References

## Appendix A — Final MVP content inventory

This list is intended to be used as a production checklist. The counts are bounded targets for the initial release, not a mandate to create filler assets.

### Player and companion

| Asset/system | MVP target | Acceptance notes |
|---|---|---|
| Mnki character | 1 production-ready rigged model | Canonical hoodie, headband, tail, pants and shoes preserved; body proportions remain consistent |
| Swan character | 1 production-ready rigged model | Canonical hood/beak/crown, hair, skirt, shoes and feathered wings preserved |
| Mnki locomotion | Idle, walk, jog, start/stop, turn, jump, land, interact, gather, fish, pet, celebrate, disappointed | Transitions should not pop; interaction state should return to locomotion safely |
| Swan companion set | Idle, follow, stop, wave, look, react, inspect, celebrate, comfort | Wing and hood silhouette must remain readable at game camera distance |
| Character switching | Not required for MVP | If approved later, input ownership, camera handoff and AI control must be explicit |
| Character portrait set | 6–10 expressions per principal character | Same palette and canonical features across UI, dialogue and marketing |

### Environment kit

| Asset family | Target range | Reuse rule |
|---|---:|---|
| Main tree variants | 6–10 silhouettes | Shared master materials, variation by size and foliage hue |
| Shrubs, flowers and ground plants | 15–25 variants | Instance densely; avoid unique materials per plant |
| Rock and ground formations | 8–12 variants | Reuse across waterfall, forest and village edges |
| Fence, path and bridge modules | 12–20 pieces | Grid-aware modular dimensions and consistent collision |
| Treehouse/home kit | 15–25 pieces | Outside pieces and interior furniture should fit one style system |
| Market stall kit | 8–12 pieces | Counters, crates, awnings, baskets and display signage |
| Farm kit | 10–16 pieces | Crop plots, watering tools, fences, compost and harvest baskets |
| Water features | 3–5 reusable modules | Pond, stream, waterfall and small decorative water areas |
| Outdoor props | 20–35 assets | Benches, lanterns, signs, pots, barrels and picnic props |

| Asset family | Target range | Reuse rule |
|---|---|---|
| VFX | 8–12 simple effects | Gather sparkle, crop growth, fishing splash, coin, quest complete, leaves and fireflies |

### Activity and content inventory

- 4–6 crop definitions with seed, growth, watering and harvest states.
- 3 domestic animal species at MVP: cow, sheep and chicken. Each has at least one idle, locomotion and interaction animation or a scoped equivalent.
- 4–6 fish definitions with different silhouettes and catch weights.
- 5–8 forage items including berries, flowers and tropical fruit.
- 6–8 named villagers, including Thronn, the best friend of Mnki and Swan.
- 10–15 short quests with varied objective patterns and clear rewards.
- 3–5 shop upgrade tiers with visible changes to stall presentation or stock capacity.
- 8–12 decorative home items that can be placed or selected through a simple furnishing interface.
- 1 opening story arc with a clear end state for the MVP.

## Appendix B — Quest template and examples

Every quest should be stored as data and follow the same schema. Avoid creating one-off quest logic inside NPC scripts.

| Field | Description |
|---|---|
| `quest_id` | Stable unique identifier; never use translated display text as an ID |
| `title` | Short readable quest name |
| `giver_id` | NPC or world object that starts the quest |
| `prerequisites` | Completed quest IDs, story flags or shop tier requirements |
| `objectives` | One or more gather, interact, deliver, visit, plant, catch, care or repair goals |
| `progress_policy` | Whether progress counts before or after acceptance |
| `rewards` | Coins, items, reputation, decoration, access or story flags |
| `completion_dialogue` | Character-specific acknowledgement of the result |
| `save_policy` | Objective progress must persist after saving and loading |

### Sample quest 1 — A Welcome Basket

**Purpose:** teach gathering, inventory, delivery and character reactions.

1. Swan suggests preparing a welcome basket for a new resident.
2. The player gathers three berries from the berry maze.
3. A small HUD counter updates after each valid pickup.
4. The player returns to the village park and interacts with the basket.
5. The basket becomes a placed world object; it is not merely removed from inventory without feedback.
6. Mnki and Swan react to the finished basket.
7. The game grants a small coin reward, friendship progress and a decorative leaf charm.
8. Save, quit, reload and verify the quest remains complete.

### Sample quest 2 — Pondside Picnic

**Purpose:** connect fishing, food gathering and the pond setting.

1. A villager plans a quiet picnic and asks for one fish and two fruits.
2. The objective tracker shows two separate requirements.
3. The player may fish immediately or gather fruit first; the order is unrestricted.
4. Invalid species/items must not satisfy the wrong objective.
5. Upon delivery, a picnic is visibly placed near the water.
6. New ambience or a small idle animation reinforces the world change.

### Sample quest 3 — The Three Guardians

**Purpose:** create a light exploration mystery and establish the Land, Water and Forest motifs.

1. The player discovers three entrance statues.
2. Each statue points toward a different region rather than providing an arbitrary fetch list.
3. One exploration observation from each region becomes a journal entry.
4. The player returns to the gate, activates the correct sequence, and opens a small garden shortcut.
5. No combat or punishment is required; clues can be revisited at any time.

## Appendix C — Sample character dialogue principles

Dialogue must show a character's values rather than repeat generic quest instructions. Keep lines short, specific to the moment, and interruptible through accessibility settings.

**Mnki — curious, playful, energetic:** - "That bridge is smaller than it looked from over there. Good thing I brought my best balance." - "One more basket, then I am absolutely taking a break. Probably." - "We made this place a little better than we found it."

**Swan — warm, perceptive, thoughtful:** - "Listen. The water sounds different when the wind turns." - "We can do it now, or tomorrow. The village will still be here." - "You noticed what they needed. That matters."

**Thronn — a reliable best friend:** - "I saved a place beside the treehouse. You two always end up standing in the path." - "The chickens are proud of their work today. Do not tell them otherwise."

These are tone samples, not final scripted dialogue. They should be reviewed against canonical character personalities before recording.

## Appendix D — Economy balancing sheet

The initial economy should be small and understandable. Start with values that allow a test loop, then tune with playtest telemetry. The figures below are prototype starting points, not final promises.

| Activity/item | Example starting value | Tuning intent |
|---|---:|---|
| Common berry | 3 coins | Reliable low-value gathering source |
| Banana | 5 coins | Common tropical fruit |
| Mango | 7 coins | Higher-value fruit with a longer route |
| Coconut | 8 coins | Higher-value gathering item |
| Common fish | 6 coins | Rewards learning the fishing timing cue |
| Egg | 4 coins | Simple animal-care output |

| Activity/item | Example starting value | Tuning intent |
|---|---|---|
| Starter seed pack | 10 coins | Affordable first farming experiment |
| Small home decoration | 30–50 coins | Near-term savings target |
| First fruit stall improvement | 100 coins | Early visible progression milestone |
| Second stall improvement | 250 coins | Medium-term progression milestone |

### Balancing rules

1. A first useful purchase should not require repetitive grinding.
2. Every core activity should be viable for progression; the player should not need to farm one optimal item endlessly.
3. Shop order rewards should provide a bonus over direct sale but not make direct sale useless.
4. Story and quest items must be protected from accidental bulk selling.
5. Add new price tiers only when new gameplay is introduced.
6. Avoid daily decay, debt pressure, animal death, or permanent crop loss in the cozy MVP.
7. Test acquisition rates and purchase timing with fresh save files; experienced developers should not be the only testers.

## Appendix E — Production asset acceptance checklist

Every imported asset must pass the following review:

- Source and license are recorded.
- Scale matches the character rig and interaction distance.
- Pivot and origin are appropriate for placement and animation.
- Materials use the shared stylized palette and approved roughness range.
- Collision is simple and intentional; decorative leaves should not become obstructive walls.
- LOD or instancing policy is documented for repeated vegetation.
- Texture resolution is justified by camera distance.
- Animation has no visible foot sliding or severe clipping at common camera angles.
- Character silhouette remains recognizable in sunlight, shadow and interior lighting.
- Imported files are named consistently and stored in the correct folder.
- A valid fallback/placeholder exists if the asset is not ready.

## Appendix F — First-playtest survey

Ask new testers to play without instruction, then collect answers to these questions:

1. What did you think the game wanted you to do first?
2. Did you know how to move, turn the camera and interact?
3. Which activity felt most satisfying, and why?
4. Did you understand how to gather an item and use or sell it?
5. Did Mnki and Swan feel distinct to you? What did you learn about them?
6. Did Swan’s companion behavior help or interrupt you?
7. Was any path, object, button or quest confusing?
8. Did you feel hurried, punished or blocked from playing at your own pace?
9. What would you most like to do during your next visit?
10. Would you want to try another build? What would need to improve first?

Record observed behavior separately from opinions. “Tester said fishing was fun” and “tester voluntarily repeated fishing three times” are different kinds of evidence.

## Appendix G — Glossary

| Term | Meaning in this GDD |
|---|---|
| Cozy loop | A low-pressure sequence of activities that provides satisfying short- and long-term progress |
| Diorama | A scene framed to feel like a small, crafted physical world |
| Interaction focus | The single target that currently receives the player’s interaction prompt |
| MVP | Minimum viable product: the smallest complete release that can test the game proposition |
| Vertical slice | A representative segment at near-target quality, proving art, code, gameplay and production pipeline together |
| Data-driven | Systems read item/quest/crop definitions from resources or data rather than duplicating logic in scripts |
| Persistent state | Information that survives saving and loading, including world changes and quest progress |
| Acceptance criterion | A testable statement that must pass before a task is considered complete |
| Scope freeze | A production boundary preventing new features from displacing launch-critical work without explicit review |
| Stretch feature | A desirable addition deferred until the MVP is stable and schedule permits it |

## Appendix H — Official and reference sources

The following sources support the technical and distribution planning. The design itself, character definitions, locations, proposed values and economy examples are original project specifications and should be playtested rather than treated as externally verified facts.

1. **Godot Engine — Godot 4.7.2 maintenance release**, 18 August 2026. Confirms the selected engine version and release notes. https://godotengine.org/article/maintenance-release-godot-4-7-2/
2. **Godot Engine — 4.7.2 stable archive entry.** Download/release archive. https://godotengine.org/download/archive/4.7.2-stable/
3. **Godot Engine — CharacterBody3D class reference.** Reference for player movement structure. https://docs.godotengine.org/en/4.7/classes/class_characterbody3d.html
4. **Godot Engine — NavigationAgent3D class reference.** Reference for companion and NPC navigation. https://docs.godotengine.org/en/4.7/classes/class_navigationagent3d.html
5. **Godot Engine — Saving games.** Reference for persistence approaches and file storage. https://docs.godotengine.org/en/4.7/tutorials/io/runtime_file_loading_and_saving.html
6. **Steamworks — Coming Soon pages.** Explains the store page, wishlist and pre-launch awareness workflow. https://partner.steamgames.com/doc/store/coming_soon
7. **Steamworks — Steam Direct fee.** Explains the per-app publishing fee and the conditions for recoupment. https://partner.steamgames.com/doc/gettingstarted/appfee
8. **Steamworks — Demos.** Guidance on creating and publishing a separate demo application. https://partner.steamgames.com/doc/store/application/demos

**Reference review date:** 22 October 2026. Confirm the latest version of external documentation before production decisions or release submissions.